GCP Professional Cloud Architect (PCA) — Cram Guide
Google Cloud Certification
Version: v1 | October 2025 Exam (v6.1) | 2026

■ Exam at a Glance

Questions	50–60 (scenario + multiple choice + multi-select)
Duration	120 minutes
Passing	~70% (Google doesn't publish exact score)
Cost	$200 USD
Validity	2 years
Prerequisite	None officially — ACE recommended
Renewal	Pass current exam before expiration

****READ THE CASE STUDIES BEFORE THE EXAM.** They are pre-published at cloud.google.com/certification/guides/professional-cloud-architect. The exam tests your ability to make decisions for these fictional companies.**

■ Domain Weightings (v6.1 — October 2025)

Domain	Weight	Key Focus
1. Designing & Planning Cloud Architecture	~24%	Requirements → GCP services, multi-region, cost estimation
2. Managing & Provisioning Infrastructure	~15%	IaC (Terraform, Deployment Manager), CI/CD, GKE
3. Designing for Security & Compliance	~18-20%	IAM, CMEK, VPC-SC, Zero Trust, audit logging
4. Analyzing & Optimizing Technical/Business	~18%	Cost optimization, Recommender API, Dataflow, sustainability
5. Managing Implementation	~11%	Traffic splitting, Config Connector, tool selection, governance
6. Ensuring Solution Reliability	~12-14%	HA/DR, RPO/RTO, SLOs, Cloud Monitoring, SRE

■ The PCA Mindset

PCA ≠ ACE. ACE tests commands (HOW). PCA tests decisions (WHY).

- Every question is a **business scenario** with constraints
- Always eliminate answers that violate stated constraints first
- Choose the **simplest solution that meets all requirements**
- GCP's managed services are almost always the right answer over self-managed

■■ Official Case Studies

Four fictional companies — know these cold:

1. Mountkirk Games

- **Type:** Mobile gaming platform
- **Key issues:** Real-time analytics, session management, player leaderboards
- **GCP services:** BigQuery, Cloud Spanner, Cloud Datastore, Compute Engine, Cloud CDN, Memorystore
- **Architecture pattern:** Real-time game analytics, globally distributed leaderboards, micro-services backend

2. Dress4Win

- **Type:** Online clothing retailer
- **Key issues:** Legacy on-premises app migration, containerization, online store
- **GCP services:** GKE (Kubernetes), Cloud SQL, Cloud Storage, VPC, Cloud Build, Migration

- **Architecture pattern:** Lift-and-shift → re-architect toward containers, CI/CD pipeline

3. TerramEarth

- **Type:** Heavy equipment manufacturer (IoT)
- **Key issues:** 500K vehicles transmitting data, bandwidth constraints, global fleet
- **GCP services:** BigQuery, Cloud IoT Core, Pub/Sub, Dataflow, Cloud Storage, Bigtable, ML
- **Architecture pattern:** IoT ingestion pipeline, edge computing, ML for predictive maintenance

4. Helix (when present)

- **Type:** Healthcare/life sciences
- **Key issues:** HIPAA compliance, data residency, secure data sharing
- **GCP services:** VPC Service Controls, Cloud Armor, CMEK, DLP API, Binary Authorization

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■ DOMAIN 1 — Designing & Planning Cloud Architecture (~24%)

Requirements → GCP Service Mapping

"Need to store unstructured files at scale, globally accessible"
→ Cloud Storage (GCS) — Standard/Multi-Regional/Regional/Nearline/Coldline/Archive tiers
"Need a managed NoSQL document database for a mobile app"
→ Firestore (Datastore mode) — for mobile/web, auto-scale
"Need a managed relational DB, PostgreSQL/MySQL compatible"
→ Cloud SQL — for standard OLTP, managed backups, HA
→ AlloyDB — for PostgreSQL at scale with AI/ML integration
→ Spanner — for global scale, strong consistency, 99.999% SLA
"Need a globally distributed relational DB with horizontal scaling"
→ Cloud Spanner — strong consistency, automatic sharding, multi-region
"Need a data warehouse for analytics at petabyte scale"
→ BigQuery — serverless, auto-scales, flat-rate vs. on-demand pricing
"Need real-time streaming analytics"
→ Pub/Sub → Dataflow (Apache Beam) → BigQuery/Cloud Storage
"Need to run containerized workloads"
→ GKE (Kubernetes) — production-grade, auto-scale, Autopilot option
→ Cloud Run — serverless containers, if stateless HTTP workloads
→ Cloud Functions — serverless functions, event-driven
"Need to deploy serverless HTTP APIs"
→ Cloud Run — containerized, scales to zero, pay per use
→ App Engine — if you need a platform instead of containers
→ Cloud Endpoints → API Gateway — for API management
"Need machine learning at scale"
→ Vertex AI — training, tuning, prediction, AutoML, endpoints
→ BigQuery ML — run ML in BigQuery using SQL
→ AI Platform (legacy) — now fully Vertex AI

Multi-Region & Global Architecture

Need	GCP Solution
Active-active multi-region	Cloud Load Balancing (global), Cloud CDN, Cloud Spanner
Active-passive failover	Cloud DNS, Traffic Director, regional managed instance groups
Disaster recovery in <1hr RTO	Cloud SQL HA, Spanner, regional persistent disks
Data archive, cheap storage	GCS Nearline/Coldline/Archive + Lifecycle policies
Global content delivery	Cloud CDN + Global External Load Balancer

Compute Options — Decision Tree

```
Is it a container?
■■■ YES — Is it stateless HTTP?
■ ■■■ YES → Cloud Run
■ ■■■ NO → GKE (Autopilot for managed)
■■■ NO — Is it a function (event-driven)?
■■■ YES → Cloud Functions
■■■ NO — Is it a full app platform?
■■■ YES → App Engine
■■■ NO — Is it a VM?
■■■ YES → Compute Engine
■■■ NO — Is it HPC/batch?
■■■ YES → Compute Engine + Spot VMs + Batch
```

Storage Decision Tree

```
Is it structured relational data?
■■■ YES — Global scale needed?
■ ■■■ YES → Cloud Spanner
■ ■■■ NO → Cloud SQL (HA for production)
■■■ NO — Is it unstructured files/blobs?
■■■ YES → Cloud Storage (choose tier by access frequency)
■■■ NO — Is it NoSQL document?
■■■ YES → Firestore/Datastore
■■■ NO — Is it wide-column?
■■■ YES → Bigtable
■■■ NO — Is it analytics/data warehouse?
■■■ YES → BigQuery
```

Pricing Tools You Must Know

Google Cloud Pricing Calculator: Estimate costs before deploying. PCA expects you to estimate.

Key pricing models:

- **On-demand** — pay per usage (BigQuery, Cloud Functions)
- **Committed Use Discounts (CUDs)** — 1 or 3 year, up to 57% savings
- **Spot VMs** — up to 91% off, preemptible, suitable for batch/ stateless
- **Sustained Use Discounts (SUDs)** — automatic discount for running VMs >25% of month
- **Flat-rate (BigQuery)** — for high-volume, predictable queries
- **Reservations** — for GKE node pools, Cloud SQL

■■ DOMAIN 2 — Managing & Provisioning Infrastructure (~15%)

Infrastructure as Code (IaC)

Terraform vs. Deployment Manager:

	Deployment Manager	Terraform
Language	YAML + Jinja2 templates	HCL (HashiCorp Configuration Language)
State	Managed by Google	You manage state file (remote: GCS)
Community	Smaller	Massive, cross-cloud
Recommended?	Legacy projects	■ Preferred for new projects

Best practice for Terraform state:

```
terraform {
backend "gcs" {
bucket = "my-terraform-state"
prefix = "prod"
}
}
```

State file should be in a versioned GCS bucket, not local.

Config Connector: Kubernetes-native way to manage GCP resources — use when you're already running GKE and want to manage GCP resources as K8s objects.

GKE — What You Must Know

GKE Autopilot: Google manages the nodes. You pay per pod resource request. No node management overhead. Recommended for production.

GKE Standard: You manage nodes, node pools, scaling. More control.

Node pools:

- System node pool (runs cluster add-ons) — at least 2 nodes
- User node pools — your workloads
- Spot node pools — cheap, preemptible (use with PodDisruptionBudgets)
- Windows node pools — for Windows containers

Workload Identity: Recommended way for GKE workloads to access GCP APIs. Binds a Kubernetes service account to a GCP service account. No keys to manage.

Multicluster:

- Anthos — Google's hybrid/multi-cloud Kubernetes management platform
- Config Sync — keep GKE clusters in sync with a central config
- Multi-cluster Ingress (MCI) — route traffic across clusters globally

CI/CD on GCP

Service	Use When
Cloud Build	Native GCP CI/CD, serverless, pay-per-minute
Jenkins / GitLab CI	Existing Jenkins/GitLab investment
ArgoCD	GitOps for GKE, declarative CD
Spinnaker	Complex multi-cloud CD pipelines
Cloud Deploy	GKE-only, managed Spinnaker-based

Networking — Core Concepts

VPC networks:

- Default network (auto-created) — avoid in production
- Custom mode VPC — you control all subnet ranges
- Shared VPC (org-level) — share VPC across projects

Subnet CIDR expansion: You can expand subnet secondary ranges without recreating the VPC. Can't shrink.

Firewall rules:

- Implied deny all ingress
- Implied allow all egress
- Always add least-privilege rules
- Use tags and service accounts, not IP ranges (IP-based rules are hard to manage)

Cloud NAT: Allow VM instances without external IPs to access the internet (for private instances). Outbound only — no inbound unsolicited connections.

Cloud DNS:

- Managed by Google, 100% SLA
- DNSSEC support
- Cloud DNS for GKE → Private DNS zone in the cluster VPC

Private Google Access: Allows VMs with only internal IPs to access Google APIs (BigQuery, GCS, etc.) without an external IP. Enable on subnet.

VPC Service Controls: Security perimeter around GCP resources. Prevents data exfiltration from services like BigQuery, Cloud Storage when accessed via approved endpoints.

■ DOMAIN 3 — Designing for Security & Compliance (~20%)

IAM — The Foundation

Principal types:

- Google Account (user)
- Service Account (application/workload)

- Google Group
- Cloud Identity domain
- Workspace domain

Resource hierarchy:

```
Organization (top)
■■■ Folder
■ ■■■ Folder / Subfolder
■ ■■■ Project
■ ■■■ Resource (VM, GCS bucket, etc.)
```

IAM policy inheritance: Organization → Folder → Project → Resource. **More restrictive policies win.**

Custom roles: Create from primitive roles (Owner/Editor/Viewer) or define at org level with fine-grained permissions. Best practice: least-privilege — use custom roles with minimum permissions.

Service Account best practices:

- Don't generate keys if you can use workload identity (GKE) or service account impersonation
- If keys needed: rotate them, store in Secret Manager
- Use service accounts per application (not shared)
- Never make a service account a Project Owner unless absolutely necessary

IAM Recommender: Shows unused permissions — use it to tighten policies.

Encryption — Know the Layers

Layer	Key Type	Who Manages Key
Storage encryption	Google-managed (AES-256)	Google
Customer-managed keys (CMEK)	Cloud KMS	You
Customer-supplied keys (CSK)	You provide key	You (key never touches Google)
Encryption in transit	TLS / VPN	Both parties
DLP API	Data inspection	You (define rules)

Cloud KMS:

- Key rings → Crypto keys → Key versions
- Crypto keys can be regional, global, or external (Cloud External Key Manager)
- Use Cloud KMS for: CMEK for GCS/BigQuery, secrets in Secret Manager, signing Docker images (Binary Authorization)

Zero Trust — Beyond the Perimeter

- **BeyondCorp** — Google's internal Zero Trust model
- **IAP (Identity-Aware Proxy)** — proxy that enforces IAM-based access to VM/web apps. No VPN needed.
- **Context-Aware Access** — restrict access based on device compliance, IP, location
- **Cloud Armor** — WAF, DDoS protection, geo-based IP restrictions

Compliance & Audit

Common frameworks tested:

- **HIPAA** — healthcare (US)
- **PCI-DSS** — payment cards
- **FedRAMP** — US government
- **GDPR** — EU data protection
- **ISO 27001** — information security management

Cloud DLP API: Inspect, classify, de-identify sensitive data (PII, credit cards, etc.)

Audit Logs:

- Admin Activity logs — always recorded, can't be disabled
- Data Access logs — Cloud Storage, BigQuery (can be expensive, often disabled by default)
- Enable based on compliance needs

Binary Authorization:

- Enforce that only signed, trusted container images are deployed to GKE

- Use Cloud KMS to sign images
- Policy modes: Allowlist (default deny) vs. Trust store

■ DOMAIN 4 — Analyzing & Optimizing (~18%)

BigQuery — Deep Dive

Architecture:

- Serverless — no infrastructure to manage
- Separates compute and storage
- Columnar storage (Capacitor — Colossus)
- Petabyte scale

Pricing:

- On-demand: $5/TB scanned (you pay for what you scan)
- Flat-rate: based on slot hours (for heavy, predictable workloads)
- Free: 1TB/month on-demand, 10GB storage, ML queries in BigQuery

Partitioning and Clustering:

Feature	Partitioning	Clustering
What it does	Splits table into logical segments	Sorts data by column(s)
Use for	Time-series data, date-based queries	High-cardinality columns
Benefits	Reduces bytes scanned, improves performance	Reduces data scanned per query
Auto-reclustering	Yes (in the background)	No — manual recluster needed

Best practice: Partition by date/timestamp, cluster by commonly filtered columns.

Materialized Views: Pre-computed results that auto-refresh. Great for frequently aggregated data.

BigQuery ML:

- Run ML models using SQL in BigQuery
- Supported: Linear regression, XGBoost, K-means, ARIMA, PCA, Matrix Factorization, etc.
- No data movement — ML happens where data lives

Dataflow — Apache Beam

What it is: Unified stream and batch processing. Serverless. Auto-scales.

Key concepts:

- **Pipelines:** Data processing job (reading → transform → write)
- **PTransforms:** Apply transforms to PCollections
- **Windowing:** Fixed, sliding, session-based windows for streaming
- **Triggers:** When to emit results (default, early, late firings)

Templates:

- Streaming templates — for production jobs
- Flex templates — custom templates for complex pipelines
- Dataflow SQL — SQL interface for beam pipelines

Use cases:

- Streaming ETL: Pub/Sub → Dataflow → BigQuery
- Log processing: GCS → Dataflow → BigQuery
- Real-time analytics

Pub/Sub — Messaging

Pub/Sub Lite: Lower cost, Zonal only (not cross-region). Use for very high throughput where cross-region isn't needed.

Pub/Sub (standard):

- At-least-once delivery (with acknowledgements)
- Exactly-once delivery (in-order, with deduplication)
- Push and pull subscriptions
- Dead-letter topics for failed messages

Ordering keys: Ensure messages with same key are delivered in order.

Cloud Logging & Monitoring

Cloud Logging:

- Log-based metrics
- Log sink to: GCS, BigQuery, Pub/Sub (for third-party SIEM)
- Log router: include/exclude filters by log name, severity

Cloud Monitoring:

- **Uptime checks:** HTTP/TCP/SSL checks from multiple locations
- **Alerting policies:** Metric thresholds, health checkers
- **SLO monitoring:** Define SLOs, burn rate alerts
- **Dashboards:** Custom dashboards with charts

SRE Practices for PCA:

- **SLIs (Service Level Indicators):** What you measure (latency, error rate, throughput)
- **SLOs (Service Level Objectives):** Target values for SLIs (99.9% availability)
- **Error budgets:** 100% - SLO = error budget. Spend it wisely.
- **Burn rate alerts:** Alert when SLO is burning faster than acceptable

Cost Optimization

Technique	Savings	When to Use
Committed Use Discounts (CUDs)	Up to 57%	Steady, predictable VMs
Spot VMs	Up to 91%	Batch, fault-tolerant workloads
Sustained Use Discounts	Up to 30%	VMs running >25% of month (automatic)
Custom machine types	Variable	Right-size to actual need
Preemptible VMs	Up to 91%	Stateless, interruption-tolerant batch
GKE Autopilot	~30-40% vs standard	Don't want to manage nodes
Cloud Run	Scales to zero	Infrequent, event-driven
GCS lifecycle policies	Archive cold data	Auto-transition Hot → Nearline → Cold → Archive
BigQuery flat-rate	For heavy query workloads	Predictable, high-volume

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■ DOMAIN 5 — Managing Implementation (~11%)

Deployment Patterns

Traffic splitting / Blue-Green / Canary:

Pattern	How It Works	Use When
Blue-Green	Full cutover at once	Complete, immediate switch
Canary	Small % traffic to new version	Test with real users before full rollout
Rolling	Gradually replace instances	No downtime, lower risk
Feature flags	Toggle features per user segment	Gradual feature rollout

Cloud Deploy: Managed delivery pipeline for GKE. Supports progressive delivery with Skaffold. Integrates with Cloud Run.

Tool Selection — When to Use What

Need	GCP Service
Deploy containers, serverless	Cloud Run
Deploy to Kubernetes	GKE + Cloud Deploy / ArgoCD
IaC for any GCP resource	Terraform
Infrastructure in YAML for GCP	Deployment Manager

Need	GCP Service
Manage GCP resources from K8s	Config Connector
CI/CD (any platform)	Cloud Build
Secrets management	Secret Manager
Orchestrate multi-step workflows	Cloud Composer (Airflow)
Event-driven serverless	Cloud Functions
APIs and backend services	Cloud Endpoints / API Gateway
CDN and global load balancing	Cloud CDN + Global External Load Balancer

API Management

Cloud Endpoints (ESPv2):

- API gateway for GKE/Cloud Run/Cloud Functions
- Extensible Service Proxy — handles auth, logging, caching
- OpenAPI spec-based

API Gateway (newer):

- Fully managed API gateway
- Quota management, API keys, monitoring
- Based on Envoy proxy

Choosing:

- Cloud Run/GKE → API Gateway or Cloud Endpoints
- Existing Apigee investment → Apigee (enterprise-grade API management)

■■ DOMAIN 6 — Ensuring Solution Reliability (~12-14%)

High Availability vs. Disaster Recovery

Level	What It Protects Against	GCP Features
Single zone	Local disk failure	Persistent Disk snapshots
Multi-zone (within region)	AZ outage	Managed instance groups (MIG) with multi-zone
Multi-region	Region outage	Global load balancing, Cloud Spanner, Cloud Storage multi-region
Cross-cloud	Provider outage	Anthos (multi-cloud Kubernetes)

RPO and RTO — Know the Difference

Metric	Definition	GCP Service/Feature
RPO (Recovery Point Objective)	How much data you can afford to lose	Backup frequency = RPO
RTO (Recovery Time Objective)	How long until service is back	Failover speed = RTO

RPO = 0: Synchronous replication (Cloud Spanner, Zonal PD, Cloud SQL HA)

RPO < 1 hour: Near-synchronous (async replica, ~minutes)

RPO = 1-24 hours: Backup-based (Cloud Storage, BigQuery snapshots)

Backup & DR Service

GCP Backup and DR Service:

- Centralized backup management for GKE, Compute Engine, Cloud SQL
- Policy-driven: schedule, retention, encryption
- Supports cross-region disaster recovery
- Backup DR service replicates to secondary region

Cloud SQL Backup:

- Automated backups: daily, configurable time window
- On-demand backups
- Point-in-time recovery (PITR) — up to 7 days (35 days with extended retention)
- Cross-region read replica for DR

GCS Backup:

- Versioning — keep old versions of objects
- Object Lifecycle Management — auto-delete or transition based on age
- Replication: CRR (Cross-region) and dual-region

GKE Reliability

Health checks:

- **Liveness probe:** Is the container alive? Restart if failing
- **Readiness probe:** Is the container ready to serve traffic?
- **Startup probe:** For slow-starting containers (disables liveness during startup)

PodDisruptionBudgets (PDBs): Ensure minimum pods available during voluntary disruptions (node upgrades, autoscaling). Combine with Spot VMs for cost + availability.

Vertical Pod Autoscaler (VPA): Recommends or automatically adjusts CPU/memory requests based on actual usage.

Horizontal Pod Autoscaler (HPA): Scales pods based on CPU/memory/custom metrics.

Cloud Monitoring — SRE Patterns

Uptime checks: Monitor HTTP/TCP endpoints from Google's global locations. Alert when check fails.

SLO monitoring:

```
# Define SLO
SLI: request_latency < 100ms (p99)
SLO: 99.5% of requests meet SLI over 30-day window
Alert: burn rate > 14.4x (1% budget burned per hour = alert before full burn)
```

Alert policies:

- Metric threshold alerts
- Multi-condition alerts (AND/OR logic)
- Alert frequency and notification channels

■ AI/ML on GCP — PCA Must-Know

Vertex AI — The Platform:

- **AutoML:** No-code model training (images, text, tabular, video)
- **Custom training:** Bring your own ML framework (TensorFlow, PyTorch, scikit-learn)
- **Vertex AI Workbench:** Jupyter-based development environment
- **Model Registry:** Centralized model versioning and deployment
- **Feature Store (Vertex AI):** Share and reuse ML features across teams

BigQuery ML — When to Use:

- Quick prototyping with SQL
- Data already in BigQuery
- Models: XGBoost, Linear regression, K-means, ARIMA
- NOT for deep learning or very large-scale training

Pre-trained APIs (AI Platform):

- Vision AI — image classification, object detection, OCR
- Video Intelligence API — scene changes, object tracking, explicit content
- Natural Language API — entity analysis, sentiment, syntax, content classification
- Translation API — translate between 100+ languages
- Speech-to-Text / Text-to-Speech

Choosing ML approach:

```
Is your data already in BigQuery?
■■■ YES — Will BQML models suffice? (linear, XGB, k-means, ARIMA)
■ ■■■ YES → BigQuery ML
■ ■■■ NO → Vertex AI Custom Training
■■■ NO — Do you need pre-built vision/language/speech?
■■■ YES → Pre-trained APIs (Vision, NLP, Speech)
■■■ NO — Do you have labeled training data?
■■■ YES → Vertex AI AutoML or custom training
■■■ NO → Consider data labeling service, then AutoML
```

■ Decision Trees — Quick Reference

Which Compute?

```
Stateless container HTTP service?
■■■ YES → Cloud Run (serverless containers, scales to zero)

Kubernetes needed?
■■■ YES → GKE (Autopilot for prod, Standard for more control)

Event-driven function?
■■■ YES → Cloud Functions (if single-purpose) or Cloud Run (if needs containers)

Full application platform?
■■■ YES → App Engine (standard or flex)

HPC / GPU workloads?
■■■ YES → Compute Engine with GPU/TPU nodes

Batch / fault-tolerant?
■■■ YES → Compute Engine Spot VMs + Managed instance group
```

Which Database?

```
Need relational?
■■■ Global scale + strong consistency + 99.999% SLA?
■ ■■■ YES → Cloud Spanner
■ ■■■ NO → Cloud SQL (HA for production)
■■■ NO — Need NoSQL?
■■■ Document (mobile/web)?
■ ■■■ YES → Firestore
■■■ Wide-column (IoT, time-series)?
■ ■■■ YES → Bigtable
■■■ In-memory cache?
■■■ YES → Memorystore (Redis/Memcached)
NO → BigQuery (analytics) or Cloud Storage (files)
```

Which Networking?

```
Need to connect on-prem to GCP?
■■■ Dedicated private connection?
■ ■■■ YES → Cloud Interconnect (Dedicated or Partner)
■ ■■■ NO → Cloud VPN (IPSec over internet)
Need global load balancing + CDN?
■■■ YES → Global External Load Balancer + Cloud CDN
Need API gateway?
■■■ YES → API Gateway or Cloud Endpoints (ESP)
Need to restrict access to GCP services?
■■■ YES → VPC Service Controls + Private Google Access
```

■ Exam Day Tips

Read the case study first — 2-3 minutes. Note constraints: budget, compliance, team skills, timeline
Eliminate first — remove answers that violate stated requirements
"Most cost-effective" → managed services + CUDs/Spot + right-sizing

"Least operational overhead" → managed services (Cloud Run, Cloud SQL, BigQuery)
"Global, highly available" → Spanner, global LB, multi-region storage
"Lift-and-shift" → Compute Engine + Migrate for Compute Engine
"Real-time analytics" → Pub/Sub → Dataflow → BigQuery
"Zero-trust" → IAP + IAM + VPC Service Controls
"Microservices on Kubernetes" → GKE + Anthos + Config Sync
"Data residency/compliance"" → VPC Service Controls + CMEK + DLP API + Binary Authorization

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■ Official Resources

- PCA Exam Guide v6.1 (PDF)
- Case Studies
- Google Cloud Architecture Center
- Qwiklabs / Google Cloud Skills Boost

Study hard. Pass the exam. ■

Last updated: June 2026